RELATORIO DE AVALIACAO E PREPARACAO - RENUMF90

Data de Processamento: 30/03/2026 as 08:56:31 ate 30/03/2026 as 08:56:45

1. CONFIGURACAO DO MODELO BIOLOGICO

Caracteristicas (Traits): PN_kg, PD_kg, PS_kg, GPD_g-dia, PE_cm, AOL_cm2, EGS_mm, MAR_%, CAR_kg-dia, PREC_SEX, IPP_dias, PROB_3P_%, STAY_%

Efeitos Fixos: GC, Fazenda, Estacao, Lote_Manejo, Regime_Alim, Tipo_Pastagem, Localidade_Bloco, Piquete, Sexo, Geracao, Raca

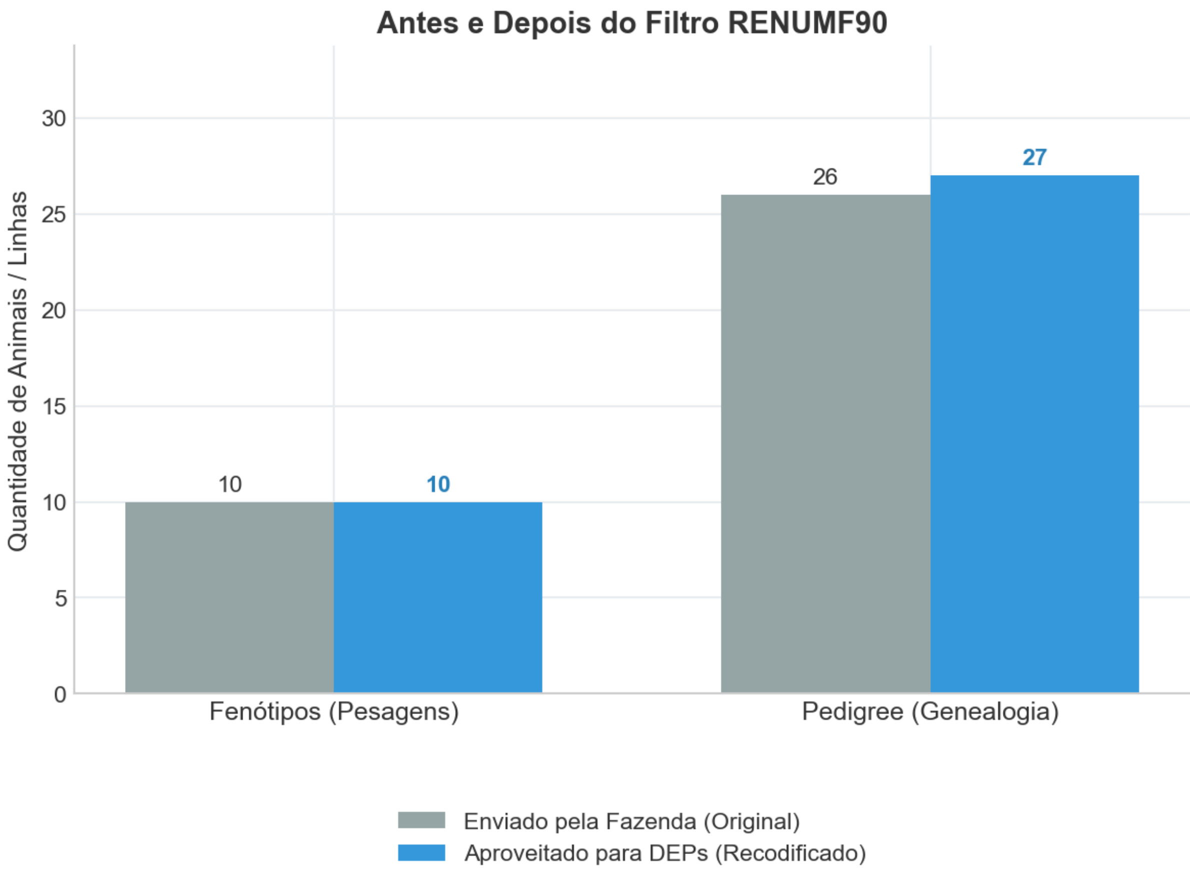
Covariaveis: itu_media, itu_dp, itu_max, itu_min, itu_leituras_criticas

2. RETENCAO E AUDITORIA DE DADOS

FENOTIPOS: Fornecidos (10) -> Aproveitados (10)

PEDIGREE: Fornecidos (26) -> Aproveitados (27)

GENOMICA: Total Analisado (26) -> Isolados/Inuteis (0)



3. SAUDE GENETICA E ALERTAS

Consanguinidade Max: N/D | Media: N/D

4. LOG TECNICO ORIGINAL (MOTOR FORTRAN)

```
========================================
[PIPELINE LINEAR]
========================================
 RENUMF90 version 1.168
* START job     Wall time: 03-30-2026  08h 56m 31s 960
 name of parameter file? renum_linear.par
 datafile:fenotipos_raw.txt
 traits:           3           4           5           6           7           8
           9          10          11          14
 R
   1.000       0.000       0.000       0.000       0.000       0.000       0.000
   0.000       0.000       0.000
   0.000       1.000       0.000       0.000       0.000       0.000       0.000
   0.000       0.000       0.000
   0.000       0.000       1.000       0.000       0.000       0.000       0.000
   0.000       0.000       0.000
   0.000       0.000       0.000       1.000       0.000       0.000       0.000
   0.000       0.000       0.000
   0.000       0.000       0.000       0.000       1.000       0.000       0.000
   0.000       0.000       0.000
   0.000       0.000       0.000       0.000       0.000       1.000       0.000
   0.000       0.000       0.000
   0.000       0.000       0.000       0.000       0.000       0.000       1.000
   0.000       0.000       0.000
   0.000       0.000       0.000       0.000       0.000       0.000       0.000
   1.000       0.000       0.000
   0.000       0.000       0.000       0.000       0.000       0.000       0.000
   0.000       1.000       0.000
   0.000       0.000       0.000       0.000       0.000       0.000       0.000
   0.000       0.000       1.000

 Processing effect  1 of type cross
 item_kind=alpha

 Processing effect  2 of type cross
 item_kind=alpha

 Processing effect  3 of type cross
 item_kind=alpha

 Processing effect  4 of type cross
 item_kind=alpha

 Processing effect  5 of type cross
 item_kind=alpha

 Processing effect  6 of type cross
 item_kind=alpha

 Processing effect  7 of type cross
 item_kind=alpha

 Processing effect  8 of type cross
 item_kind=alpha

 Processing effect  9 of type cross
 item_kind=alpha

 Processing effect 10 of type cross
 item_kind=alpha

 Processing effect 11 of type cross
 item_kind=alpha
```

```
 Processing effect 12 of type cov

 Processing effect 13 of type cov

 Processing effect 14 of type cov

 Processing effect 15 of type cov

 Processing effect 16 of type cov

 Processing effect 17 of type cross
 item_kind=alpha
 pedigree file name  "pedigree_raw.txt"
 positions of animal, sire, dam, alternate dam, yob, and group     1     2     3     0     0     0     0
 SNP file name  "genotipos_raw.txt"

 Maximum size of character fields: 20

 Maximum size of record (max_string_readline): 8000

 Maximum number of fields for input file (max_field_readline): 100

 Pedigree search method (ped_search): convention

 Order of pedigree animals (animal_order): default

 Order of UPG (upg_order): default

 Missing observation code (missing): -999

  guessed number of lines           10

 Using prime hash function
 hash tables for effects set up
 first 3 lines of the data file (up to 700 characters)
     1 122 35 210 450 695.6521739130435 38 85.5 4.2 3.5 -0.5 0 -999 28.21 0 1 1 1 4 1 1 20240322 2 114.3 75.37 6.82 87.31
-0.9371417658180198 133 1 1 1 20200101
     2 105 38 225 480 739.1304347826086 40 92.1 5.1 4.2 0.12 0 -999 72.98 0 1 1 2 1 2 1 20240226 38 114.3 77.56 5.64 86.99
0.8794964850096355 145 1 1 1 20200101
     3 141 32 195 410 623.1884057971015 36 78.4 3.8 3.1 -0.85 0 -999 83.38 0 1 2 3 1 3 1 20240611 41 114.3 68.62 5.72 78.72
-1.2598741478086302 72 1 1 1 20200101
 read           10  records in   1.5625000E-02  seconds
 ------------------------------
Processing effect group    1
 table with            6  elements sorted
 added count
 Effect group            1  of column            1  with            6  levels
 table expanded from        10000  to        10000  records
 Effect group            1  done in   0.4531250      seconds
 ------------------------------
Processing effect group    2
 table with            1  elements sorted
 added count
 Effect group            2  of column            1  with            1  levels
 table expanded from        10000  to        10000  records
 Effect group            2  done in   0.4375000      seconds
 ------------------------------
Processing effect group    3
 table with            2  elements sorted
 added count
 Effect group            3  of column            1  with            2  levels
 table expanded from        10000  to        10000  records
 Effect group            3  done in   0.4062500      seconds
 ------------------------------
Processing effect group    4
```

```
 table with           5  elements sorted
 added count
 Effect group           4  of column           1  with           5  levels
 table expanded from        10000  to        10000  records
 Effect group           4  done in   0.4843750      seconds
 ------------------------------
Processing effect group    5
 table with           2  elements sorted
 added count
 Effect group           5  of column           1  with           2  levels
 table expanded from        10000  to        10000  records
 Effect group           5  done in   0.6406250      seconds
 ------------------------------
Processing effect group    6
 table with           4  elements sorted
 added count
 Effect group           6  of column           1  with           4  levels
 table expanded from        10000  to        10000  records
 Effect group           6  done in   0.5781250      seconds
 ------------------------------
Processing effect group    7
 table with           2  elements sorted
 added count
 Effect group           7  of column           1  with           2  levels
 table expanded from        10000  to        10000  records
 Effect group           7  done in   0.5000000      seconds
 ------------------------------
Processing effect group    8
 table with           9  elements sorted
 added count
 Effect group           8  of column           1  with           9  levels
 table expanded from        10000  to        10000  records
 Effect group           8  done in   0.4687500      seconds
 ------------------------------
Processing effect group    9
 table with           1  elements sorted
 added count
 Effect group           9  of column           1  with           1  levels
 table expanded from        10000  to        10000  records
 Effect group           9  done in   0.4062500      seconds
 ------------------------------
Processing effect group   10
 table with           1  elements sorted
 added count
 Effect group          10  of column           1  with           1  levels
 table expanded from        10000  to        10000  records
 Effect group          10  done in   0.4218750      seconds
 ------------------------------
Processing effect group   11
 table with           1  elements sorted
 added count
 Effect group          11  of column           1  with           1  levels
 table expanded from        10000  to        10000  records
 Effect group          11  done in   0.5156250      seconds
 ------------------------------
Processing effect group   12
 ------------------------------
Processing effect group   13
 ------------------------------
Processing effect group   14
 ------------------------------
Processing effect group   15
 ------------------------------
Processing effect group   16
 ------------------------------
Processing effect group   17
```

added count
Effect group 17 of column 1 with 10 levels
Effect group 17 done in 0.0000000E+00 seconds
wrote statistics in file "renf90.tables"
reading data file for basic statistics

Basic statistics for input data (missing value code is '-999')

Pos	Min	Max	Mean	SD	N
3	32.000	40.000	35.900	2.6013	10
4	195.00	240.00	215.70	14.545	10
5	410.00	510.00	456.00	34.222	10
6	623.19	782.61	696.52	58.036	10
7	35.000	42.000	38.200	2.2998	10
8	78.400	98.000	87.190	6.5314	10
9	3.8000	6.2000	4.6700	0.81247	10
10	3.1000	4.8000	3.8100	0.56657	10
11	-1.2000	0.45000	-0.25400	0.50724	10
14	28.210	83.380	60.337	17.025	10
25	68.300	77.910	74.659	4.3673	10
26	5.5600	6.8200	5.8100	0.36830	10
27	77.290	88.790	84.859	4.7581	10
28	-1.3066	0.98301	-0.59605E-08	1.0000	10
29	66.000	151.00	123.00	37.425	10

Correlation matrix

	3	4	5	6	7	8	9	10	11	14	25	26	27	28	29
3	1.00	0.99	0.98	0.95	0.93	0.97	0.97	0.98	0.21	-0.19	0.54	-0.26	0.48	0.64	0.53
4	0.99	1.00	0.99	0.96	0.94	0.98	0.98	0.99	0.17	-0.20	0.55	-0.29	0.49	0.65	0.54
5	0.98	0.99	1.00	0.99	0.97	0.99	0.97	0.98	0.13	-0.20	0.52	-0.24	0.46	0.60	0.52
6	0.95	0.96	0.99	1.00	0.98	0.98	0.94	0.95	0.09	-0.20	0.49	-0.20	0.44	0.55	0.49
7	0.93	0.94	0.97	0.98	1.00	0.98	0.95	0.96	0.13	-0.20	0.38	-0.23	0.34	0.46	0.38
8	0.97	0.98	0.99	0.98	0.98	1.00	0.97	0.98	0.18	-0.21	0.46	-0.27	0.40	0.55	0.45
9	0.97	0.98	0.97	0.94	0.95	0.97	1.00	0.99	0.14	-0.16	0.48	-0.35	0.42	0.61	0.48
10	0.98	0.99	0.98	0.95	0.96	0.98	0.99	1.00	0.22	-0.16	0.47	-0.35	0.41	0.60	0.45
11	0.21	0.17	0.13	0.09	0.13	0.18	0.14	0.22	1.00	0.18	-0.08	-0.11	-0.14	-0.01	-0.10
14	-0.19	-0.20	-0.20	-0.20	-0.20	-0.21	-0.16	-0.16	0.18	1.00	-0.21	-0.61	-0.32	0.03	-0.22
25	0.54	0.55	0.52	0.49	0.38	0.46	0.48	0.47	-0.08	-0.21	1.00	-0.02	0.99	0.92	1.00
26	-0.26	-0.29	-0.24	-0.20	-0.23	-0.27	-0.35	-0.35	-0.11	-0.61	-0.02	1.00	0.10	-0.40	0.02
27	0.48	0.49	0.46	0.44	0.34	0.40	0.42	0.41	-0.14	-0.32	0.99	0.10	1.00	0.85	0.99
28	0.64	0.65	0.60	0.55	0.46	0.55	0.61	0.60	-0.01	0.03	0.92	-0.40	0.85	1.00	0.90
29	0.53	0.54	0.52	0.49	0.38	0.45	0.48	0.45	-0.10	-0.22	1.00	0.02	0.99	0.90	1.00

Counts of nonzero values (order as above)

```
        10        10        10        10        10        10        10        10        10        10        10        10
 10        10        10
        10        10        10        10        10        10        10        10        10        10        10        10
 10        10        10
        10        10        10        10        10        10        10        10        10        10        10        10
 10        10        10
        10        10        10        10        10        10        10        10        10        10        10        10
 10        10        10
        10        10        10        10        10        10        10        10        10        10        10        10
 10        10        10
        10        10        10        10        10        10        10        10        10        10        10        10
 10        10        10
        10        10        10        10        10        10        10        10        10        10        10        10
 10        10        10
        10        10        10        10        10        10        10        10        10        10        10        10
 10        10        10
        10        10        10        10        10        10        10        10        10        10        10        10
 10        10        10
        10        10        10        10        10        10        10        10        10        10        10        10
 10        10        10
        10        10        10        10        10        10        10        10        10        10        10        10
 10        10        10
        10        10        10        10        10        10        10        10        10        10        10        10
```

```
  10        10        10
         10        10        10        10        10        10        10        10        10        10        10        10
  10        10        10
         10        10        10        10        10        10        10        10        10        10        10        10
  10        10        10
         10        10        10        10        10        10        10        10        10        10        10        10
  10        10        10
 basic_statistics done in   0.0000000E+00  seconds

 random effect with SNPs 17
 type: animal
 file: genotipos_raw.txt
 read SNPs          26  records
 Effect group           17  of column            1  with           27  levels
 add_snp done in   0.0000000E+00  seconds
 reorder_animal done in   0.0000000E+00  seconds

 random effect  17
 type:animal
renumber_animal, start    Wall time: 03-30-2026  08h 56m 38s 810
 opened output pedigree file "renadd17.ped"
 first 3 lines of the file (up to 700 characters)
    1 3012 4055
    2 3005 4210
    3 3044 4102
 load_animals time   0.0000000E+00  seconds
 read           26  pedigree records
 loaded            0  parent(s) in round            1
 load_animals time   0.0000000E+00  seconds
 read           26  pedigree records

 Pedigree checks
compute_inbreeding time, start    Wall time: 03-30-2026  08h 56m 38s 815

 Computations for inbreeding coefficients
actual inbreeding computation, start    Wall time: 03-30-2026  08h 56m 38s 815
actual inbreeding computation, done     Wall time: 03-30-2026  08h 56m 38s 815
 Tiny negative value will be replaced with 0 considered as numerical error.
 Wrote inbreeding file "renf90.inb" with original id

 Inbreeding statistics:
 the maximum inbreeding coefficient    =  0.0000
 average inbreeding for inbred animals =     NaN   n = 0
                       for all animals =  0.0000   n = 27
compute_inbreeding time, done     Wall time: 03-30-2026  08h 56m 38s 819
 loop writing renadd, time   0.0000000E+00  seconds

 Number of animals with records                  =           10
 Number of animals with genotypes                =           26
 Number of animals with records or genotypes     =           27
 Number of animals with genotypes and no records =           17
 Number of parents without records or genotypes  =            0
 Total number of animals                         =           27
renumber_animal, end      Wall time: 03-30-2026  08h 56m 38s 823

 Wrote cross reference IDs for SNP file "genotipos_raw.txt_XrefID"
 write_SNP_Xref done in   0.0000000E+00  seconds
 write_map done in   0.0000000E+00  seconds

 Wrote parameter file "renf90.par"
 write renf90.par done in   0.0000000E+00  seconds
 writing renf90.dat
 Wrote renumbered data "renf90.dat" 10 records
 write renf90.dat done in   0.0000000E+00  seconds

 Number of records NOT present in pedigree file: 1
```

```
 Wrote field information "renf90.fields" for 27 fields in data
 save_renum_table done in   1.5625000E-02  seconds
* END  job     Wall time: 03-30-2026  08h 56m 38s 837


========================================
[PIPELINE CATEGÓRICO (LIMIAR)]
========================================
 RENUMF90 version 1.168
* START job     Wall time: 03-30-2026  08h 56m 38s 925
 name of parameter file? renum_categ.par
 datafile:fenotipos_raw.txt
 traits:          12          13          15
 R
   1.000       0.000       0.000
   0.000       1.000       0.000
   0.000       0.000       1.000

 Processing effect  1 of type cross
 item_kind=alpha

 Processing effect  2 of type cross
 item_kind=alpha

 Processing effect  3 of type cross
 item_kind=alpha

 Processing effect  4 of type cross
 item_kind=alpha

 Processing effect  5 of type cross
 item_kind=alpha

 Processing effect  6 of type cross
 item_kind=alpha

 Processing effect  7 of type cross
 item_kind=alpha

 Processing effect  8 of type cross
 item_kind=alpha

 Processing effect  9 of type cross
 item_kind=alpha

 Processing effect 10 of type cross
 item_kind=alpha

 Processing effect 11 of type cross
 item_kind=alpha

 Processing effect 12 of type cov

 Processing effect 13 of type cov

 Processing effect 14 of type cov

 Processing effect 15 of type cov

 Processing effect 16 of type cov

 Processing effect 17 of type cross
 item_kind=alpha
 pedigree file name  "pedigree_raw.txt"
 positions of animal, sire, dam, alternate dam, yob, and group     1     2     3     0     0     0     0
 SNP file name  "genotipos_raw.txt"
```

```
 Maximum size of character fields: 20

 Maximum size of record (max_string_readline): 8000

 Maximum number of fields for input file (max_field_readline): 100

 Pedigree search method (ped_search): convention

 Order of pedigree animals (animal_order): default

 Order of UPG (upg_order): default

 Missing observation code (missing): -999

  guessed number of lines           10

 Using prime hash function
 hash tables for effects set up
 first 3 lines of the data file (up to 700 characters)
     1 122 35 210 450 695.6521739130435 38 85.5 4.2 3.5 -0.5 0 -999 28.21 0 1 1 1 4 1 1 20240322 2 114.3 75.37 6.82 87.31
-0.9371417658180198 133 1 1 1 20200101
     2 105 38 225 480 739.1304347826086 40 92.1 5.1 4.2 0.12 0 -999 72.98 0 1 1 2 1 2 1 20240226 38 114.3 77.56 5.64 86.99
0.8794964850096355 145 1 1 1 20200101
     3 141 32 195 410 623.1884057971015 36 78.4 3.8 3.1 -0.85 0 -999 83.38 0 1 2 3 1 3 1 20240611 41 114.3 68.62 5.72 78.72
-1.2598741478086302 72 1 1 1 20200101
 read           10  records in   0.0000000E+00  seconds
 ------------------------------
Processing effect group    1
 table with            6  elements sorted
 added count
 Effect group            1  of column            1  with            6  levels
 table expanded from        10000  to        10000  records
 Effect group            1  done in   0.4218750      seconds
 ------------------------------
Processing effect group    2
 table with            1  elements sorted
 added count
 Effect group            2  of column            1  with            1  levels
 table expanded from        10000  to        10000  records
 Effect group            2  done in   0.5000000      seconds
 ------------------------------
Processing effect group    3
 table with            2  elements sorted
 added count
 Effect group            3  of column            1  with            2  levels
 table expanded from        10000  to        10000  records
 Effect group            3  done in   0.5156250      seconds
 ------------------------------
Processing effect group    4
 table with            5  elements sorted
 added count
 Effect group            4  of column            1  with            5  levels
 table expanded from        10000  to        10000  records
 Effect group            4  done in   0.4531250      seconds
 ------------------------------
Processing effect group    5
 table with            2  elements sorted
 added count
 Effect group            5  of column            1  with            2  levels
 table expanded from        10000  to        10000  records
 Effect group            5  done in   0.4687500      seconds
 ------------------------------
Processing effect group    6
 table with            4  elements sorted
 added count
```

```
 Effect group            6  of column            1  with            4  levels
 table expanded from        10000  to        10000  records
 Effect group            6  done in   0.4062500      seconds
 ------------------------------
Processing effect group    7
 table with            2  elements sorted
 added count
 Effect group            7  of column            1  with            2  levels
 table expanded from        10000  to        10000  records
 Effect group            7  done in   0.4375000      seconds
 ------------------------------
Processing effect group    8
 table with            9  elements sorted
 added count
 Effect group            8  of column            1  with            9  levels
 table expanded from        10000  to        10000  records
 Effect group            8  done in   0.5468750      seconds
 ------------------------------
Processing effect group    9
 table with            1  elements sorted
 added count
 Effect group            9  of column            1  with            1  levels
 table expanded from        10000  to        10000  records
 Effect group            9  done in   0.4531250      seconds
 ------------------------------
Processing effect group   10
 table with            1  elements sorted
 added count
 Effect group           10  of column            1  with            1  levels
 table expanded from        10000  to        10000  records
 Effect group           10  done in   0.4531250      seconds
 ------------------------------
Processing effect group   11
 table with            1  elements sorted
 added count
 Effect group           11  of column            1  with            1  levels
 table expanded from        10000  to        10000  records
 Effect group           11  done in   0.4375000      seconds
 ------------------------------
Processing effect group   12
 ------------------------------
Processing effect group   13
 ------------------------------
Processing effect group   14
 ------------------------------
Processing effect group   15
 ------------------------------
Processing effect group   16
 ------------------------------
Processing effect group   17
 added count
 Effect group           17  of column            1  with           10  levels
 Effect group           17  done in   0.0000000E+00  seconds
 wrote statistics in file "renf90.tables"
 reading data file for basic statistics

 Basic statistics for input data  (missing value code is '-999')
 Pos  Min         Max         Mean        SD                 N
   12    0.0000     0.10000E-09  0.0000      0.0000          10
   13   0.10000E+11 0.10000E-09  0.0000      0.0000           0
   15    0.0000     0.10000E-09  0.0000      0.0000          10
   25    68.300      77.910      74.659      4.3673          10
   26    5.5600      6.8200      5.8100     0.36830          10
   27    77.290      88.790      84.859      4.7581          10
   28   -1.3066     0.98301    -0.59605E-08  1.0000          10
   29    66.000      151.00      123.00      37.425          10
```

```
 Correlation matrix
       12    13    15    25    26    27    28    29
 12    NaN  0.00   NaN   NaN   NaN   NaN   NaN   NaN
 13   0.00  0.00  0.00  0.00  0.00  0.00  0.00  0.00
 15    NaN  0.00   NaN   NaN   NaN   NaN   NaN   NaN
 25    NaN  0.00   NaN  1.00 -0.02  0.99  0.92  1.00
 26    NaN  0.00   NaN -0.02  1.00  0.10 -0.40  0.02
 27    NaN  0.00   NaN  0.99  0.10  1.00  0.85  0.99
 28    NaN  0.00   NaN  0.92 -0.40  0.85  1.00  0.90
 29    NaN  0.00   NaN  1.00  0.02  0.99  0.90  1.00

 Counts of nonzero values (order as above)
         10         0        10        10        10        10        10        10
          0         0         0         0         0         0         0         0
         10         0        10        10        10        10        10        10
         10         0        10        10        10        10        10        10
         10         0        10        10        10        10        10        10
         10         0        10        10        10        10        10        10
         10         0        10        10        10        10        10        10
         10         0        10        10        10        10        10        10
  Adjusting residuals for empty subclasses
 *** Warning *** no records for           2 -th trait!!!
 *** The variance component is not estimable!!!
 basic_statistics done in   0.0000000E+00  seconds

 random effect with SNPs 17
 type: animal
 file: genotipos_raw.txt
 read SNPs          26  records
 Effect group           17  of column            1  with           27  levels
 add_snp done in   0.0000000E+00  seconds
 reorder_animal done in   0.0000000E+00  seconds

 random effect  17
 type:animal
renumber_animal, start    Wall time: 03-30-2026  08h 56m 45s 513
 opened output pedigree file "renadd17.ped"
 first 3 lines of the file (up to 700 characters)
    1 3012 4055
    2 3005 4210
    3 3044 4102
 load_animals time   0.0000000E+00  seconds
 read           26  pedigree records
 loaded            0  parent(s) in round           1
 load_animals time   0.0000000E+00  seconds
 read           26  pedigree records

 Pedigree checks
compute_inbreeding time, start    Wall time: 03-30-2026  08h 56m 45s 518

 Computations for inbreeding coefficients
actual inbreeding computation, start    Wall time: 03-30-2026  08h 56m 45s 518
actual inbreeding computation, done     Wall time: 03-30-2026  08h 56m 45s 519
 Tiny negative value will be replaced with 0 considered as numerical error.
 Wrote inbreeding file "renf90.inb" with original id

 Inbreeding statistics:
 the maximum inbreeding coefficient    =  0.0000
 average inbreeding for inbred animals =     NaN   n = 0
                      for all animals =  0.0000   n = 27
compute_inbreeding time, done     Wall time: 03-30-2026  08h 56m 45s 522
 loop writing renadd, time   0.0000000E+00  seconds

 Number of animals with records                  =           10
 Number of animals with genotypes                =           26
```

```
 Number of animals with records or genotypes     =           27
 Number of animals with genotypes and no records =           17
 Number of parents without records or genotypes  =            0
 Total number of animals                         =           27
renumber_animal, end      Wall time: 03-30-2026  08h 56m 45s 525

 Wrote cross reference IDs for SNP file "genotipos_raw.txt_XrefID"
 write_SNP_Xref done in   0.0000000E+00  seconds
 write_map done in   0.0000000E+00  seconds

 Wrote parameter file "renf90.par"
 write renf90.par done in   0.0000000E+00  seconds
 writing renf90.dat
 Wrote renumbered data "renf90.dat" 10 records
 write renf90.dat done in   0.0000000E+00  seconds

 Number of records NOT present in pedigree file: 1
 Wrote field information "renf90.fields" for 20 fields in data
 save_renum_table done in   1.5625000E-02  seconds
* END  job     Wall time: 03-30-2026  08h 56m 45s 540
```